Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is a ray infinite in one direction? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A ray has one endpoint and extends forever in one direction. The answer is yes.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Does a line extend infinitely in both directions? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A line has no endpoints and continues forever in both directions.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

How many points are needed to define a unique line? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Two distinct points determine exactly one line.**

Correct answer: **2**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A reflection maps segment AB to A′B′. Is AB congruent to A′B′? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Reflections are rigid motions. They preserve length, so the segments are congruent.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 40° angle is rotated about a point. What is the measure of the image angle? Student's answer: 80°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Rotations preserve angle measure. The image angle is still 40°.**

Correct answer: **40°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Does a dilation preserve distances? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A dilation with scale factor not equal to 1 changes distances, so it is not a rigid motion.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$\triangle ABC \cong \triangle DEF$, and $\angle A = 50°$. What is $\angle D$? Student's answer: 130°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Corresponding angles of congruent triangles are congruent. $\angle D = 50°$.**

Correct answer: **50°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$\triangle PQR \cong \triangle XYZ$, and PQ = 7. What is XY? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student doubled the length. Corresponding sides are equal, so XY = 7.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In $\triangle ABC \cong \triangle DEF$ with A ↔ D, B ↔ E, and C ↔ F, which side corresponds to AC? Student's answer: DE

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **AC uses the vertices A and C, which match D and F. The corresponding side is DF.**

Correct answer: DF

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two triangles have sides 3, 4, and 5 in both. Are they congruent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **All three pairs of sides are congruent, so the triangles are congruent by SSS.**

Correct answer: **Yes, by SSS**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two triangles have angles 50° and 60° with a side of 4 between those angles in both. Which criterion proves congruence? Student's answer: AAA

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The side is between the two angles, so the criterion is ASA. AAA does not prove congruence.**

Correct answer: **ASA**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is SSA a valid criterion for proving triangles congruent? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **SSA can describe two different triangles, so it is not a valid congruence criterion.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles are vertical angles. One measures 70°. What is the other? Student's answer: 110°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Vertical angles are congruent. The other angle is 70°.**

Correct answer: **70°**

Find the Error

Grade 10 · G.CO.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two parallel lines are cut by a transversal. An alternate interior angle is 65°. What is the other?
Student's answer: 115°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Alternate interior angles are congruent, so the other angle is 65°.**

Correct answer: **65°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A point is on the perpendicular bisector of a segment. Is it equidistant from the endpoints?
Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Every point on the perpendicular bisector is equidistant from the endpoints.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has angles of 40° and 75°. What is the third angle? Student's answer: 75°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student copied one angle. 180 − (40 + 75) = 65°.**

Correct answer: **65°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has a base of 12. What is the length of its midsegment parallel to the base? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The midsegment parallel to the third side is half that side. Its length is 6.**

Correct answer: 6

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An exterior angle of a triangle is 115°. One remote interior angle is 50°. What is the other remote interior angle? Student's answer: 165°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the angles. 115 - 50 = 65°.**

Correct answer: **65°**

Find the Error

Grade 10 · G.CO.11

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a parallelogram, one angle is 110°. What is the opposite angle? Student's answer: 70°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Opposite angles of a parallelogram are congruent. The opposite angle is 110°.**

Correct answer: **110°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The diagonals of a parallelogram meet at O. If AC = 14, what is AO? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The diagonals bisect each other, so AO is half of AC. AO = 7.**

Correct answer: 7

Find the Error

Grade 10 · G.CO.11

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a parallelogram, one angle is 110°. What is an adjacent angle? Student's answer: 110°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Consecutive angles of a parallelogram add to 180°. The adjacent angle is 70°.**

Correct answer: **70°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A dilation with scale factor 3 maps a side of length 4 to a new side. What is the new length?
Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 3 to 4. Multiply: 4 × 3 = 12.**

Correct answer: **12**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are two triangles similar if two pairs of angles are congruent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Two pairs of congruent angles guarantee similarity by AA.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a similar pair, a side of length 3 in the small triangle matches a side of 6 in the large triangle. A side of 4 in the small triangle is what in the large? Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The scale factor is 6 ÷ 3 = 2. So 4 × 2 = 8.**

Correct answer: **8**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Similar triangles have AB = 4, DE = 6, and BC = 8. What is EF if EF corresponds to BC? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **4/6 = 8/x, so x = 12. The student added the sides.**

Correct answer: **12**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line parallel to BC crosses AB at D and AC at E. AD = 3, DB = 6, and AE = 4. What is EC? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **AD/DB = AE/EC, so 3/6 = 4/EC. Then EC = 8.**

Correct answer: 8

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a right triangle, the altitude to the hypotenuse splits it into segments of 4 and 9. What is the altitude? Student's answer: 13

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The altitude is the geometric mean of the segments: $\sqrt{(4 \times 9)} = 6$.**

Correct answer: 6

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Triangles have two sides and the included angle congruent. Which method proves congruence?
Student's answer: SSA

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The angle is between the two sides, so the correct criterion is SAS.**

Correct answer: SAS

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Similar triangles have sides 5 and 10 matching, and a side of 14 in the second triangle matches a side of x in the first. Find x. Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **5/10 = x/14, so x = 7.**

Correct answer: **7**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In similar triangles, AB = 3 and DE = 6, and AC = 5. What is DF? Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The scale factor is 6 ÷ 3 = 2. DF = 2 × 5 = 10.**

Correct answer: **10**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a right triangle, the opposite side to angle A is 3 and the hypotenuse is 5. What is sin A?
Student's answer: 4/5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **4/5 is cos of the other angle. sin A = 3/5.**

Correct answer: **3/5**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a right triangle, the adjacent side to angle B is 8 and the hypotenuse is 17. What is cos B?
Student's answer: 15/17

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **15/17 uses the opposite side. cos B = 8/17.**

Correct answer: **8/17**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is tan 45°? Student's answer: $\sqrt{2}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **In a 45-45-90 triangle, the opposite and adjacent legs are equal, so tan 45° = 1.**

Correct answer: 1

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has legs 3 and 4. What is the hypotenuse? Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the legs. $3^2 + 4^2 = 25$, so the hypotenuse is 5.**

Correct answer: 5

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has opposite side 5 and hypotenuse 10 for angle A. What is angle A? Student's answer: 60°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **sin A = 5/10 = 1/2, so A = 30°.**

Correct answer: **30°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 10-foot ladder makes a 60° angle with the ground. How high does it reach? Student's answer: 5 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used cosine. Height = 10 sin 60° = 5√3, about 8.66 feet.**

Correct answer: **5√3, about 8.66 feet**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A central angle measures 80°. What is the inscribed angle that intercepts the same arc?
Student's answer: 80°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **An inscribed angle is half the central angle that intercepts the same arc. 80 ÷ 2 = 40°.**

Correct answer: 40°

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An inscribed angle measures 35°. What is the intercepted arc? Student's answer: 35°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The intercepted arc is twice the inscribed angle, so 2 × 35 = 70°.**

Correct answer: 70°

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An inscribed angle is drawn with its sides on a diameter. What is its measure? Student's answer: 180°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The intercepted arc is a semicircle, which is 180°. The inscribed angle is half of that, so it is 90°.**

Correct answer: 90°

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the arc length of a 90° arc in a circle with radius 6. Student's answer: 6π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found the length of a full half circle. (90/360) × 12π = 3π, about 9.42.**

Correct answer: **3π, about 9.42**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a 60° sector in a circle with radius 3. Student's answer: 9π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found the area of the full circle. (60/360) × 9π = 1.5π, about 4.71.**

Correct answer: **1.5π, about 4.71**

Find the Error

Grade 10 · G.C.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the arc length of a 120° arc in a circle with radius 9. Student's answer: 12π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 2π × 9 × 2/3 incorrectly. (120/360) × 18π = 6π.**

Correct answer: **6π**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle has vertices A(0, 0), B(4, 0), C(4, 3), and D(0, 3). What is the length of diagonal AC?
Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the side lengths. AC = $\sqrt{(4^2 + 3^2)} = 5$.**

Correct answer: 5

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the slope of the line through A(1, 2) and B(5, 10)? Student's answer: 1/2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student flipped the ratio. The slope is 8 ÷ 4 = 2.**

Correct answer: 2

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are the lines $y = 2x + 1$ and $y = 2x - 3$ parallel? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Both lines have slope 2 and different intercepts, so they are parallel.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the midpoint of segment AB with A(0, 0) and B(10, 0). Student's answer: (4, 0)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a subtraction error. The midpoint is (5, 0).**

Correct answer: **(5, 0)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the point 1/3 of the way from A(0, 0) to B(12, 0). Student's answer: (3, 0)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided 12 by 4. 1/3 of 12 is 4, so the point is (4, 0).**

Correct answer: **(4, 0)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the point that divides A(2, 0) to B(8, 0) in a 2:1 ratio, measured from A. Student's answer: (5, 0)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 1/2 of the distance. 2/3 of 6 is 4, so the point is (6, 0).**

Correct answer: **(6, 0)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle has vertices (0, 0), (5, 0), (5, 2), and (0, 2). What is its perimeter? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted each side once. The perimeter is 2(5 + 2) = 14.**

Correct answer: **14**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has vertices (0, 0), (6, 0), and (0, 8). What is its area? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the legs. Area = 1/2 × 6 × 8 = 24.**

Correct answer: 24

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A square has vertices (1, 1), (4, 1), (4, 4), and (1, 4). What is its area? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: 12 is the perimeter. The side is 3, so the area is 9.

Correct answer: 9

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the circumference of a circle with radius 4. Student's answer: 4π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to multiply by 2. $C = 2\pi(4) = 8\pi$, about 25.13.**

Correct answer: **8π, about 25.13**

Find the Error

Grade 10 · G.GMD.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cylinder with radius 3 and height 5. Student's answer: 15π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to square the radius. V = π(9)(5) = 45π, about 141.37.**

Correct answer: **45π, about 141.37**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cone with radius 3 and height 6. Student's answer: 54π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot the 1/3. $V = (1/3)\pi(9)(6) = 18\pi$, about 56.55.**

Correct answer: **18π, about 56.55**

Find the Error

Grade 10 · G.GMD.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box is 4 by 5 by 6 units. What is its volume? Student's answer: 15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied only two dimensions. V = 4 × 5 × 6 = 120.**

Correct answer: **120**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A cylinder has volume 200π and radius 5. What is its height? Student's answer: 40

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided by 5 instead of 25. 200π = π(25)h, so h = 8.**

Correct answer: 8

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A sphere has volume 36π. What is its radius? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided the volume by π and stopped. $36 = (4/3)r^3$ gives $r^3 = 27$, so $r = 3$.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which solid best models a soup can? Student's answer: cone

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A soup can has two circular bases and straight sides, so a cylinder is the best model.**

Correct answer: **Cylinder**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which solid best models a basketball? Student's answer: cylinder

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A basketball is round in every direction, so a sphere is the best model.**

Correct answer: **Sphere**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangular garden is 12 feet by 7 feet and needs fencing around its edge. How much fencing is needed? Student's answer: 19 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added only one length and one width. Perimeter = 2(12 + 7) = 38 feet.**

Correct answer: **38 feet**